

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

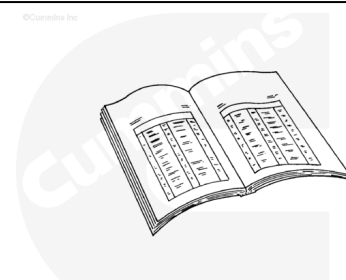
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Drain the oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>).
- Remove the lubricating oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html>).
- Remove the oil pan adapter. Refer to Procedure 007-027 (</qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html>).
- Remove the oil pump, if necessary. Refer to Procedure 007-031 (</qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html>).
- Remove the block stiffener plate.

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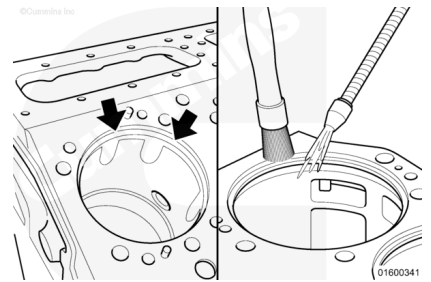


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Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use a scraper or a fibrous, abrasive pad, such as Scotch-Brite™ 7477, or equivalent, and solvent. Clean the top deck of the block. Do **not** allow any dirt into the cylinder. The surface **must** be free of dirt, oil, and gasket material, but does **not** have to look like new metal.

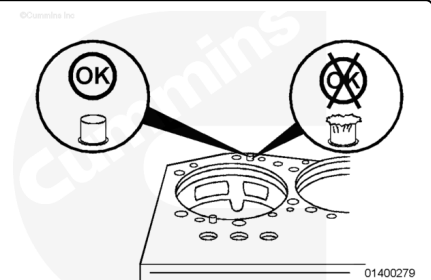
Note : The counterbore ledge **must** be clean and dry. The sealant used in assembly will **not** dry if oil is present.

Check the top surface of the block for wear. If fretting damage is present in an area where a head gasket seal ring or a grommet makes contact, the surface **must** be repaired. Refer to the Alternative Repair Manual, Bulletin 3379035.

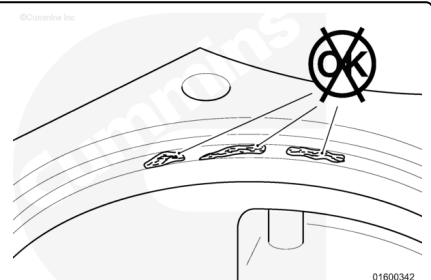
Fretting damage in any other area is acceptable **only** if it does **not** change the protrusion measurement of the counterbore or liner.

Inspect the dowel pins for damage.

Use the dowel pin extractor, Part Number ST-1134, or equivalent, to remove the pins if they are damaged.

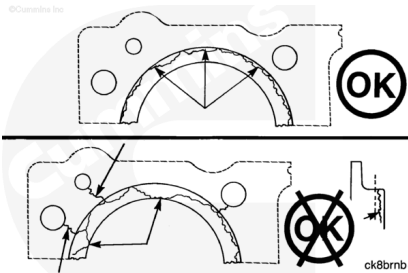


Inspect the counterbore ledge for pitted areas. The ledge **must** be machined if pitting is present.

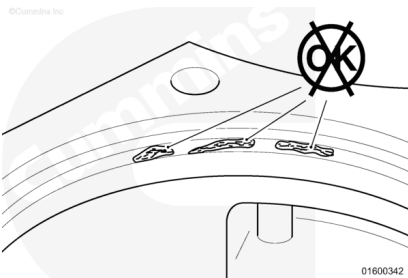


Check for cracks.

Use a crack detection kit, Part Number 337532 or equivalent, and check the counter bore ledges for cracks. Circumferential cracks are acceptable if they do **not** extend more than 1/2 of the distance across the ledge.



Cracks that extend into a capscrew hole are **not** acceptable. The block can be used if a capscrew hole thread insert is installed in the capscrew hole. See the Alternative Repair Manual, Bulletin 3379035.

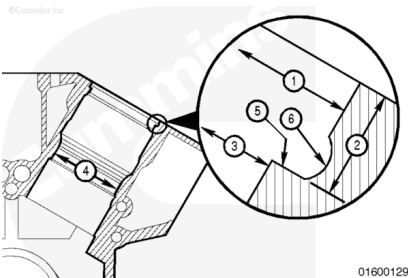


Inspect the counterbore

To inspect the block properly, the counterbore design **must** be determined.

Terms used are:

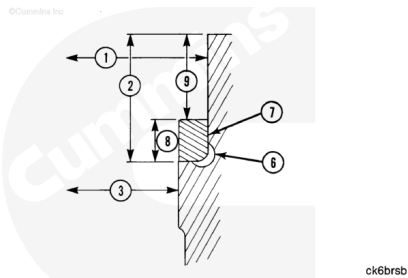
1. Upper counterbore inside diameter
2. Counterbore depth
3. Lower counterbore inside diameter
4. Packing ring bore
5. Counterbore ledge
6. Counterbore radius.



Measure

Counterbore Ring - Thick Deck Block Design, most K38 engines with engine serial number first greater than 33100193 contain this design. All K50 engines contain this design

(1) The actual size depends on whether standard size or oversize liners were installed.



millimeters	Oversize [inches]	inches
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190.284 to 190.335	[standard]	7.4915 to 7.4935
190.538 to 190.589	[0.010]	7.5015 to 7.5035
190.792 to 190.843	[0.020]	7.5115 to 7.5135
191.046 to 191.097	[0.030]	7.5215 to 7.5235
191.300 to 191.351	[0.040]	7.5315 to 7.5335

(2) The actual size depends on whether standard size or oversize liners were installed. The size also depends on whether the block was cut for shims

millimeters	Oversize [inches]	inches
18.26 to 18.31	[standard]	0.718 to 0.720
18.52 to 18.57	[0.010]	0.728 to 0.730
18.77 to 18.82	[0.020]	0.738 to 0.740
19.02 to 19.07	[0.030]	0.748 to 0.750
19.28 to 19.33	[0.040]	0.758 to 0.760

The depth can be a maximum of 3.39 mm [0.133 inch] greater if the block was machined previously for shims and oversize liners.

(3) engine serial number first lower than 33101028

millimeters		inches
181.74 to 181.79		7.155 to 7.157

Note : Some engines with an engine serial number first lower than 33101028 can have the above (3) greater than the specification. The cylinder block can be used again only if the correct press fit between the cylinder liner and dimensions (1) is obtained.

(6) This design contains a double undercut radius.

(7) This design contains a counterbore ring that functions as a load distribution washer (8).

millimeters	Oversize [inches]	inches
5.055 to 5.067	[standard]	0.1990 to 0.1995
5.105 to 5.118	[0.002]	0.2010 to 0.2015
5.156 to 5.169	[0.004]	0.2030 to 0.2035

(9) This depth depends on the size of the liner. Liner protrusion **must** be 0.152 to 0.203 mm [0.006 to 0.008 in].

millimeters	Oversize [inches]	inches
13.16 to 13.23	[standard]	0.519 to 0.521
13.41 to 13.49	[0.010]	0.529 to 0.531
13.67 to 13.74	[0.020]	0.539 to 0.541
13.92 to 14.00	[0.030]	0.549 to 0.551
14.17 to 14.25	[0.040]	0.559 to 0.561

If the upper counterbore is **not** within standard specifications, the block **must** be machined for oversize liners.

If the block is larger than the specification for the largest oversize, a repair sleeve can be installed. See the Alternative Repair Manual Bulletin, 3379035.

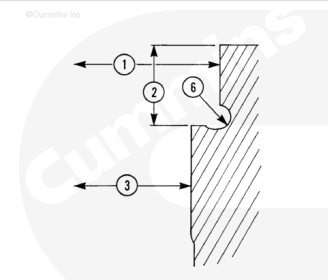
If **only** the counterbore depth (2) is **not** within specifications, the block **must** be machined for shims. Before machining the block for shims, check to see if the protrusion can be corrected by installing an oversize counterbore ring.

⚠ CAUTION ⚠

Do not attempt to machine blocks of this design for a counterbore ring as it will cause the block to break.

Thin Deck Block - Double Undercut Radius Design, some K38 standby generator set engines with an engine serial number first higher than 33100193 contain this design. No K50 engines contain this design.

(1) The actual size depends on whether the standard size or oversize liners were installed.



ck4brgb

millimeters	Oversize [inches]	inches
190.284 to 190.335	[standard]	7.4915 to 7.4935
190.538 to 190.589	[0.010]	7.5015 to 7.5035
190.792 to 190.843	[0.020]	7.5115 to 7.5135
191.046 to 191.097	[0.030]	7.5215 to 7.5235
191.300 to 191.351	[0.040]	7.5315 to 7.5335

(2) These specifications are required for a liner protrusion of 0.152 to 0.203 mm [0.006 to 0.008 inch]. The protrusion specification changed since blocks this design were introduced.

The actual size depends on whether the standard size or the oversize liners were installed, and if the block was machined for shims.

millimeters	Oversize [inches]	inches
13.16 to 13.23	[standard]	0.519 to 0.521
13.41 to 13.49	[0.010]	0.529 to 0.531
13.67 to 13.74	[0.020]	0.539 to 0.541
13.92 to 14.00	[0.030]	0.549 to 0.551
14.17 to 14.25	[0.040]	0.559 to 0.561

(3) Some blocks of this design can have dimension (3) larger than the specification. The cylinder block can be used again **only** if the correct press fit between the liner and dimension (1) is obtained.

millimeters		inches
181.74 to 181.79		7.155 to 7.157

(6) This design contains a double undercut radius.

If the upper counterbore is **not** with standard specifications, the block **must** be machined for shims.

If **only** the counterbore depth (2) is **not** within specifications, the block **must** be machined for shims. See the procedure, Cylinder - Cut Counterbore Depth for shims.

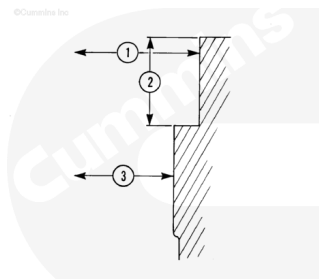
If the block is larger than the specification for the largest oversize liner, a repair sleeve can be installed. See the Alternative Repair Manual Bulletin, 3379035.

It is recommended that a block of this design be used **only** for K38 standby generator set applications.

⚠ CAUTION ⚠

Do not attempt to machine blocks of this design for a counterbore ring as it will cause the clock to break.

Thin Deck Block Design, most K38 engines with an engine serial number first lower than 33100193 contain this design. No K50 engines contain this design.



(1) The actual size depends on whether the standard size or oversize liners were installed.

millimeters	Oversize [inches]	inches
190.284 to 190.335	[standard]	7.4915 to 7.4935
190.538 to 190.589	[0.010]	7.5015 to 7.5035
190.792 to 190.843	[0.020]	7.5115 to 7.5135
191.046 to 191.097	[0.030]	7.5215 to 7.5235
191.300 to 191.351	[0.040]	7.5315 to 7.5335

Note : These specifications are those required for a liner protrusion of 0.152 to 0.203 mm [0.006 to 0.008 inch]. The protrusion specification has changed since blocks of this design were produced.

(2) The actual size depends on whether the standard size or the oversize liners were installed, and if the block was machined for shims.

millimeters	Oversize [inches]	inches
13.16 to 13.23	[standard]	0.519 to 0.521
13.41 to 13.49	[0.010]	0.529 to 0.531
13.67 to 13.74	[0.020]	0.539 to 0.541
13.92 to 14.00	[0.030]	0.549 to 0.551
14.17 to 14.25	[0.040]	0.559 to 0.561

(3) Some blocks of this design can have dimension (3) larger than the specification. The cylinder block can be used again **only** if the correct press fit between the liner and dimension (1) is obtained.

millimeters	inches
181.74 to 181.79	7.155 to 7.157

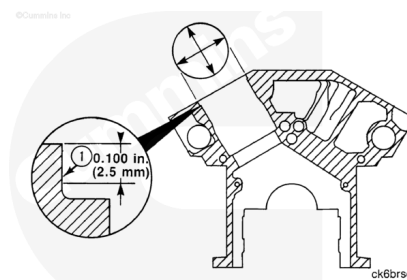
This design does **not** contain the double undercut radius.

It is recommended that a block of this design be used **only** for K38 standby generator set applications, or those applications requiring a rated power of 675 kW [900 bhp] or less.

Measure the inside diameter of the upper counterbore (1). The point of measurement **must** be within 2.5 mm [0.100 inch] from the top surface of the block.

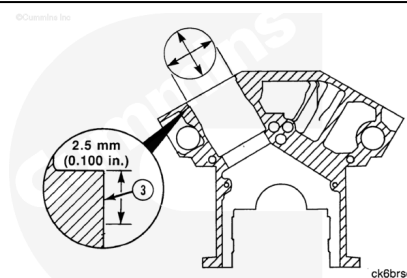
The diameter of the upper counterbore **must** be round within 0.025 mm [0.001 inch]. If the measurement is **not** within specification, determine if the block can be machined for oversize liners. See Procedure cylinder block - Cut for Oversize Liners.

Note : The upper counterbore **must** be no more than 0.025 mm [0.001 inch] larger than the cylinder liner flange, or no more than 0.076 mm [0.003 inch] smaller than the cylinder liner flange.

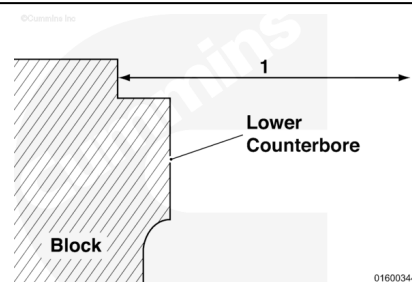


Measure the inside diameter of the lower counterbore (3). The point of measurement **must** be within 2.5 mm [0.100 inch] from the top of the counterbore ledge. Measure at two locations, 90 degrees apart

The diameter of the lower counterbore **must** be within 0.025 mm [0.001 inch].



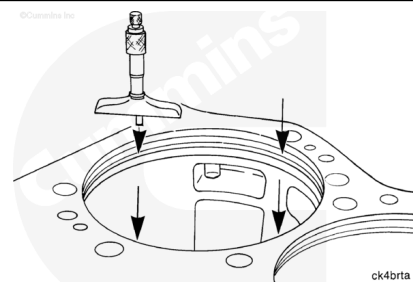
If the diameter of the lower counterbore is larger than the specification, the block can be used **only** if the outside diameter of the liner flange is 0.025 mm [0.001 inch] larger than the inside diameter of the upper counterbore. If there is **not** a press fit between the liner and the upper counterbore, the block **must** be machined for oversize liners (dimension 1).



⚠ CAUTION ⚠

Be sure the micrometer contacts the flat surface of the counterbore ledge. It must not touch the counterbore face.

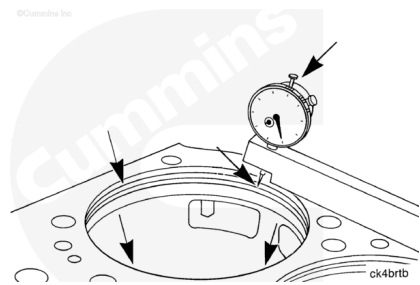
Use a depth micrometer. Measure the counterbore depth at the four places shown. The four measurements **must not** vary more than 0.025 mm [0.001 inch] . If the measurement are **not** in accordance with the specifications, the counterbore ledge **must** be machined. In this case, see the Procedure 001-058 (/qs3/pubsys2/xml/en/procedures/28/28-001-058.html).



⚠ CAUTION ⚠

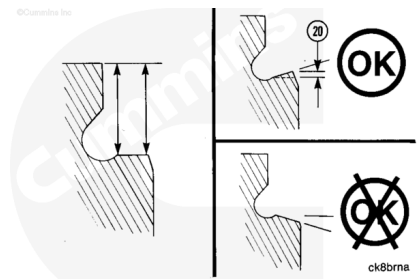
Be sure the indicator does not contact the counterbore face (radius) on a block that does not have a double undercut.

Use Part Number ST-537 Dial Depth Gauge. Measure the angle of the counterbore ledge at no less than four locations around the ledge. One measurement at each location **must** be as close as possible to the counterbore edge.



It is acceptable if the counterbore ledge has an angle no greater than described as follows:

- If the distance measured near the counterbore radius (face) is the same or no more than 0.036 mm [0.0014 inch] farther than the distance measured near the counterbore edge, the counterbore ledge is acceptable.
- The counterbore ledge angle is **not** acceptable if the distance near the counterbore edge is farther than the distance measured near the counterbore radius (face), In this case, the ledge **must** be machined. Reference Alternative Repair Manual, Bulletin 3379035 for repairing the packing ring bore with a bushing.

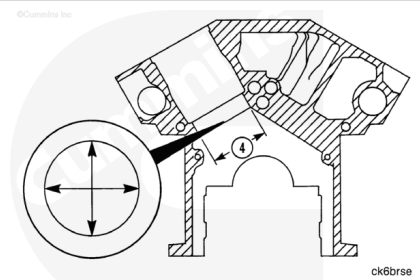


Measure the packing ring bore.

Closely inspect the chamfer at the top of the packing ring bore.

Excessive pitting of the chamfer or the ring bore is **not** acceptable. See the Alternative Repair Manual, Bulletin 3379035 for repairing the packing ring bore with a bushing.

Measure the diameter of the packing ring bore.

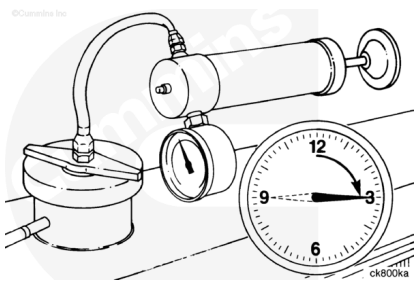


mm		in
177.34	MIN	6.982
177.39	MAX	6.984

Leak Test

Pressurize the engine cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html).

Apply the air pressure 15 minutes before inspecting the cylinder liner seats. Make sure the system is holding air pressure before beginning inspection.



Measurements

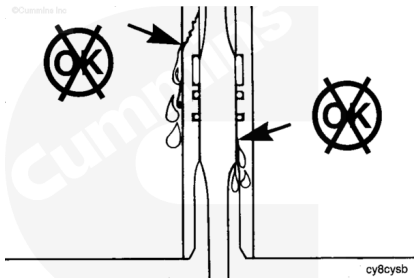
	kpa	psi
Air Pressure	138	20

Inspect the outside diameter of the cylinder liners and the area below the cylinder liner seats in the cylinder block for coolant leaks.

If a leak is found, remove the cylinder liner(s). Inspect the sealing ring and cylinder liner. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html).

If a leak is found in the cylinder liner seat. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html).

Inspect the cylinder block liner counterbore area.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the block stiffener plate.
- Install the oil pump, if removed. Refer to Procedure 007-031 (/qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html).
- Install the oil suction and transfer tubes. Refer to Procedure 007-035 (/qs3/pubsys2/xml/en/procedures/28/28-007-035.html).
- Install the oil pan adapter. Refer to Procedure 007-027 (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html).

- Install the oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html>).
- Fill the engine with clean engine oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>).
- Operate the engine until it reaches a temperature of 80°C [176°F].
- Check for coolant and lubricating oil leaks.

Last Modified: 09-Jan-2007
